

Mapping the Ghost Town of Van Houten, Colfax Co., N.M.

Kai Brennan

Independent Researcher

NRA Whittington Center Adventure Camp

Fort Collins, Colorado

kai.j.brennan@gmail.com — (303) 997-8058

*Abstract—****

*Index Terms—****

I. INTRODUCTION

II. RELATED WORK

As Van Houten was a relatively small company town, and would not have needed a town plan or cadastral survey, there are few contemporary maps of the town as a high-detail primary subject. However, the town has been captured on numerous other large-scale maps for both population-related sources and technical ones. The mining operations are also more clearly detailed on several key maps and surveys.

A. Technical Sources

Technical maps of Van Houten first include the internal mine maps that were produced during operations. These are detailed in [1]- [3]. These are of great use for understanding both the detailed internal structure of the room and pillar mines, as well as the early sections that were subject to longwall mining. They are equally useful for town mapping and having an idea of the foundations and commune layout for those included on the maps.

Additionally, overall reports on the Raton Coal Field, such as in [4] & [5], provide a more big picture layout for the mine's adits and drifts as of 1913. The company maps in [6] & [7] also show a larger picture of the mining operations while being more specific to the town.

Finally, modern surveys such as [8]- [10] have explored the area and highlighted the remaining traces of certain husks of the town and mining operations.

B. Cultural Sources

Census Enumeration districts from 1940 and 1950, in [11] & [12] show the general layout of the buildings in the town, and the placement of some important structures like the post office. These same layouts are also depicted in a more detailed manner in [4] & [5].

Complementarily, recent explorers have recorded certain situational details for landmarks or made simple maps in [13] & [14]. Invaluably, past residents have as recounted useful information for mapping, as in [15].

C. Tangential Sources

There is a wealth of additional information contained within other archives and databases, from photographs to aerial imagery.

Contemporary sources that have been highly useful to triangulating completely demolished buildings are in [16]- [18]. Additionally, global railroads may come to be understood through [19].

Beyond this, modern exploration may be credited for many of the discoveries and advancements in the mapping of Van Houten. The Youtube videos in [20]- [22] have detailed aerial imagery that is instrumental in stitching together various foundations that are not visible from other map datasets. Beyond the aforementioned sources, much work has been put into the exploration of Van Houten by the counselor staff of the NRA Whittington Center Adventure Camp, who have freely shared points and discoveries among themselves. Notably prolific people include Tucker Barraclough, Markus Hecker, among others.

III. DATA

Many accessible sources of high quality aerial imagery were used in piecing together a working map of the town of Van Houten. These helped specifically for the foundations that are visible today that overwrote those shown on old company maps in [1]- [3] (particularly in Monte Hill), and for places where there were no detailed maps (Helen's Hood).

Each service provided disparate advantages, and the most significant contributors are enumerated below:

- 1) Apple Maps—This service had the highest resolution aerial imagery of the town, and was helpful to find both foundations and things usually obscured by foliage.
- 2) Caltopo—This web tool provided many useful layers, including relief shading, historical maps, and topographic maps.
- 3) On X—This map was the highest quality summer imagery, and made the stone foundations stand out.
- 4) Google Earth — The tool for viewing satellite imagery over time allowed one to see the mining reclamation projects, in [23] & [24], that took place. It also allowed for one to reverse ruins in certain areas to see them in a more complete state.

IV. METHODS

Faithfully reconstructing the layout of Van Houten's residential structures proved difficult due to the need to pick a certain reference time frame and an objective metric for foundation inclusion. Van Houten was built over itself in some areas (most notably Monte Hill). This layering made it difficult to determine which foundations should be depicted on a map, as newer structures intersect or replace older ones.

Additionally, certain photographs or other tangential sources may have depicted buildings that are impossible to triangulate within the town's layout with any degree of confidence given the current data. These had to be omitted to avoid leading researchers or explorers astray (though notable omissions are detailed in §8).

To create a comprehensive and intelligible map, each foundation had to meet a set of criteria (non-significant structures were omitted (i.e. outhouses, garages, cellars, wells, etc.)):

- 1) All foundations with known visible, physical remnants were mapped with primary importance.
- 2) For foundations no longer visible:
 - a) The most recent known structure at a given spot took precedence, effectively overwriting any earlier buildings it replaced or covered.
 - b) Inclusion on the map required some form of physical or corroborative evidence that was consistent and detailed enough to confidently approximate the structure's shape and position onto a satellite base map. This evidence included appearance on historical mine maps, depressions on shaded relief imagery, satellite data, artifact scatter, road alignment, or triangulation via known positions in period photographs.

This created a highly useful map that shows a researcher or explorer the most recent structure at any given location. While the buildings may not have all existed at the same time, they represent what was last present, and thus what would still be found regarding debris, ruins, or artifacts today. This novel map style has been named an "Ephemeral Recency Map" (ERM).

Mine mapping was far easier than foundational mapping, for there exist internal mine maps that were drawn with high accuracy in [1]- [3]. Additionally, the surveys presented in [4]- [10] provide data relevant in locating the historic adits. On top of this, significant exploration has found other adits. Knowledge of adit locations was pieced together with bigger picture mine schemes to create the overlay displayed on the map.

Through these standards, a project on QGIS was compiled to hold the necessary layers for display and printing.

V. TOWN PLAN

Explain how I sectioned it into communes*

A. Monte Hill

Monte Hill is the hardest commune within VH to piece together for a map. Many of the foundations of Monte Hill are

*.png

Fig. 1. Commune Sections Assembly Map.

still visible and make a regular grid pattern of mostly regularly sized foundations, yet it does not line up perfectly with [1] (although it seems to correspond in some areas). Not many photos exist of this region, and it appears that the buildings may have experienced significant change between the years. The flat "hill" was conducive to expansion or re-development, and the low income of the Montenegrin population likely contributed to this. The large building of "Shops" was on Monte Hill. Monte Hill was partially covered in [10]. As stated earlier, the most recent foundations (i.e. those still visible) are the ones that appear on the map.

B. Industrial Quarter

The Industrial Quarter contains many well preserved and significant foundations, including the mule barn, the tipple, the powerhouse, and the bathhouse, as well as "Grandma Spaghetti's" standing house. The archaeological survey in [10] contains a partial overview of this region, and many photos in [16] show the no longer present structures of this commune.

C. West Common

West Common is one of the more difficult communes. Much of the buildings have been destroyed and are spread out. Important structures include the boarding house, the doctor's office, the schoolhouse, and the company store. These structures are still visible on aerial imagery, and photos from the tipple in [16] show the layout of the area. There are some buildings that may lie under the road or have been lost elsewhere.

D. High Five

High Five and the accompanying shed are mostly visible today, and their layout is clearly explained in [15]. Each house was square shaped and had an attached rose garden. Many images from this commune are present in [16].

E. Helen's Hood

Helen's Hood contains Helen Roupas' standing house, Soup Bone, the company well, and Nestor's Grave. Most of the foundations in this commune are in good shape and are very visible in the drone footage from [20] & [21], as well as on aerial imagery. Images of the commune exist in [16], but they are slightly difficult to piece together exactly.

F. Negro Canyon

There are six plainly visible foundations that have been mapped in Negro Canyon. However, due to the great construction that has occurred in this region from the dam and the fire-mine access road, one should question if others have been lost. No granular maps or images exist of the Negro Canyon settlements. That said, the maps in [4] & [5] depict a comparatively small number of buildings in this commune, so it is possible they have all been discovered.

G. Sunshine Hill

All of the foundations on Sunshine Hill are visible from aerial imagery and may be compared with overview photos from [16] to confirm the positioning of the structures. Due to the hill's geography, little expansion or change occurred in this area.

H. East Common

East Common contains the clubhouse and the feet of the tippie that serviced mine number five by way of the funicular. The foundation of the clubhouse and several collapsed buildings are visible, but it is possible that other foundations were covered under the clubhouse or otherwise lost. The best sources for this area may be found in [16].

I. Coon Town

Coon Town was the easiest commune of VH to map. All but one of the original foundations are plainly visible on both Apple Maps and Google Earth. Additionally, all of the foundations are depicted in [3], and directly correspond with those visible today. The commune's geographical positioning between two hillsides limited expansion, and gave it a highly regular distribution of sizes and placements, as well as little re-development. The most north-eastern house foundation (numbered 237 in [3]) lies under the modern road, with no remnants.

J. Greek Town

Greek Town was also comparatively easy to survey. It is fully depicted in [3]. While only a few foundations are visible today (including the Willow Post Office), the depressions where the demolished foundations once stood are plainly visible on relief shading layers from Caltopo. The layout of this town consists of three lines of square and rectangular houses running diagonally southeast. Two of the lines are south of the modern road, and the more southern of them is offset slightly to the east. Despite the intensive demolitions, the foundation depressions are similar today to what was depicted in [3], so one may infer there was not much re-development in Greek Town.

K. Cunico Town

Cunico Town is relatively well preserved, and through the visible foundations (concentrated at the southern end of the commune), one can position the map in [3] to triangulate the position of the other demolished houses. There are contemporary elevated photographs in [16] that reaffirm their positioning. It seems that this town did not experience significant expansion or redevelopment.

L. Graveyard Canyon

Graveyard Canyon did not have any standing structures.

VI. MINE LAYOUT

A. Mine No. 1

B. Mine No. 2

C. Mine No. 3

D. Mine No. 4

E. Mine No. 5

F. Mine No. 6

G. Backwoods Mine

H. Devil's Kitchen

VII. RAILWAYS

VIII. QUESTIONABLE AREAS

IX. FUTURE RESEARCH

REFERENCES

- [1] St. Louis, Rocky Mountain and Pacific Company. (n.d.). Van Houten No. 4 coal mine map [Map No. 2450, PDF]. Unpublished manuscript.
- [2] St. Louis, Rocky Mountain and Pacific Company. (n.d.). Van Houten No. 4 coal mine map [Map No. 2452, PDF]. Unpublished manuscript.
- [3] St. Louis, Rocky Mountain and Pacific Company. (n.d.). Van Houten No. 6 coal mine map [Map No. 2451, PDF]. Unpublished manuscript.
- [4] Lee, W. T. (1922). Raton-Brilliant-Koehler folio, New Mexico-Colorado (Folios of the Geologic Atlas 214) [Plates for Quad 1: Areal Geology, Economic Geology, and Topography]. U.S. Geological Survey. <https://doi.org/10.3133/gf214>
- [5] Lee, W. T. (1924). Coal resources of the Raton coal field, Colfax County, New Mexico (U.S. Geological Survey Bulletin 752) [Plate 1]. U.S. Geological Survey. <https://doi.org/10.3133/b752>
- [6] St. Louis, Rocky Mountain and Pacific Company. (1921, September 1). General outcrop map, Raton, Colfax County, New Mexico [Map No. 2404, PDF]. Unpublished manuscript.
- [7] Raton coal field, Colfax County, New Mexico [Map No. 2233, PDF]. (1943). Unpublished manuscript.

- [8] Nickelson, H. B. (1979). Reclamation evaluation for OSM, Raton Field, Willow-Van Houten Mines Nos. 1 through 6 [NM Mine File No. 2308]. NMBMMR. Unpublished manuscript.
- [9] Nickelson, H. B. (1979). Reclamation evaluation for OSM, Raton Field, Willow-Van Houten Mines Nos. 1 through 6 [Map No. 5184]. New Mexico Bureau of Geology and Mineral Resources. Unpublished manuscript.
- [10] Oakes, Y. R. (1987). Archaeological Survey of Mine Lands in the Raton Coal Field for the Abandoned Mine Land Bureau. Laboratory of Anthropology, Abandoned Mine Land Bureau. <https://www.academia.edu/116997177/>
- [11] Department of Commerce, Bureau of the Census. (1940). 1940 Census Enumeration District maps – New Mexico – Colfax County – ED 4-1 to 4-47 [Maps]. National Archives. <https://catalog.archives.gov/id/5835351>
- [12] Department of Commerce, Bureau of the Census. (1950). 1950 Census Enumeration District Maps - New Mexico, Colfax County - ED 4-1 to 43 [Map]. National Archives. <https://catalog.archives.gov/id/23928668>
- [13] Ballew, M. et al. (1973). Van Houten History. NRA Whittington Center.
- [14] Barraclough, T. (n.d.). Van Houten Tour [Unpublished manuscript].
- [15] Yaksich, G. ("Judo"). (2022, August 4). Interview by T. Barraclough [Transcript]. Unpublished manuscript.
- [16] Raton Museum. (n.d.). Photographs related to the Town of Van Houten, Colfax County, New Mexico [Photograph collection]. Raton Museum Archives, Raton, NM.
- [17] Koehler, A. E. et al. (1915). New Mexico, the Land of Opportunity: Official Souvenir of the State of New Mexico at Panama California Exposition San Diego 1915. A. E. Koehler.
- [18] General catalogue of maps: Gretchen K. Hoffman. (n.d.). Inactive Coal Mines and Georeferenced Maps Raton Basin, Colfax County, New Mexico (Open-file Report 572). New Mexico Bureau of Geology and Mineral Resources. https://geoinfo.nmt.edu/publications/openfile/downloads/500-599/572/OFR_572.pdf
- [19] Railroads: Scott, G. R. (1986). Historic trail maps of the Raton and Springer 30' x 60' quadrangles, New Mexico and Colorado (IMAP 1641) [Plate 1]. U.S. Geological Survey. <https://doi.org/10.3133/i1641>
- [20] Far From Ordinary. (2023, April 26). Camping at a GHOST TOWN (The Story of Van Houten) [ep 123] [Video]. Youtube. <https://youtu.be/j5PQHfKRQTW?si=eac7vBH2kGUzHFjH>
- [21] DJ Newell. (2021, November 4). Van Houten Ghost Town [Video]. Youtube. https://youtu.be/NpjQ0gp2B1c?si=bN_NF_JstM2OmIR4
- [22] Boots and Bonnets with Chris & Christy. (2023, August 10). EXPLORING VAN HOUTON NEW MEXICO [Video]. Youtube. <https://youtu.be/Lw7fyv3nnIA?si=hjcUaBi8qG7nrcRD>
- [23] New Mexico Energy, Minerals, and Natural Resources Department. (n.d.). Abandoned Mine Land Program Projects. New Mexico Energy, Minerals, and Natural Resources Department. <https://www.emnrd.nm.gov/mmd/abandoned-mine-land-program/projects/>
- [24] Mine fire: Lohmann, P. (2024, October 22). State extinguishes Gallup-area mine fires burning since at least 2011. Source NM. <https://sourcenm.com/briefs/state-extinguishes-gallup-area-mine-fires-burning-since-at-least-2011/>

*.png

Fig. 2. Commune Sectional Map with Foundations Marked.

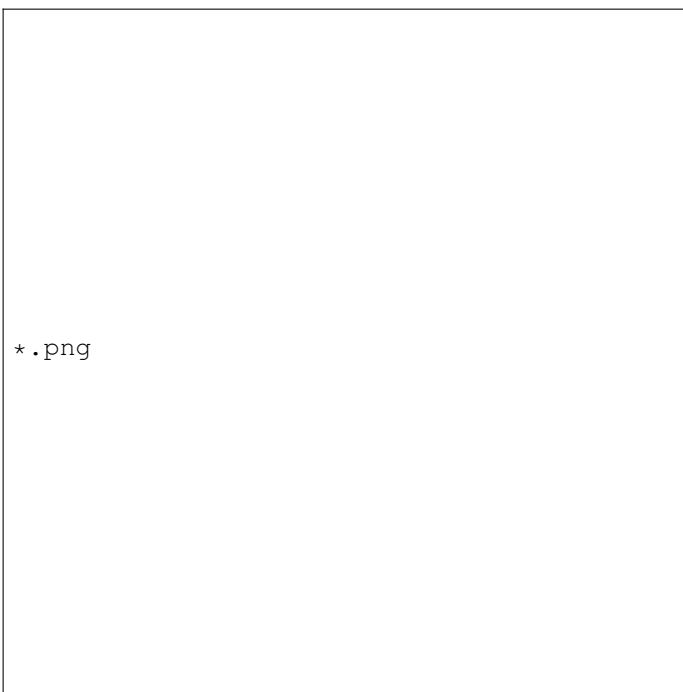


Fig. 3. Mine Sections Assembly Map.

*.png

Fig. 4. Mine Sectional Map with Main Adits and Drifts Depicted.